

ScholarAgent

**Evidence-Grounded Agentic RAG for Scholarship Discovery
and Eligibility Verification**

Technical Project Report

Authors: Tivdzua Lubem Noah and Gisele Wiykiynyuy
Institution: Innopolis University
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A reproducible and failure-aware scholarship discovery system

Abstract

Scholarship discovery requires more than retrieving relevant webpages. Applicants must interpret eligibility rules, distinguish active from expired opportunities, and verify that recommendations are supported by authoritative evidence. ScholarAgent addresses these requirements through an evidence-grounded Agentic Retrieval-Augmented Generation (RAG) workflow that combines official-source scholarship records, lexical and semantic retrieval, deterministic eligibility screening, citation verification, bounded repair, deterministic fallback, and safe abstention.

The system was evaluated on a frozen held-out corpus containing six scholarship programmes and 24 benchmark cases. BM25 achieved perfect Recall@3, Mean Reciprocal Rank, and Top-1 accuracy on this controlled corpus. The deterministic eligibility engine achieved perfect accuracy and macro F1 across four balanced eligibility classes. Conventional single-pass RAG failed citation verification on every positive case. The complete Agentic RAG workflow achieved citation-safe completion on all positive cases, although every successful completion required deterministic verified fallback rather than a successful TinyLlama repair. This improvement increased mean latency from 140.251 seconds to 293.064 seconds and doubled generation calls.

The results show that explicit verification and deterministic recovery can improve trustworthiness in a small local-agent setting. They also show that the strongest contribution of the agent architecture was controlled decision flow and safe failure handling rather than unconstrained language-model autonomy.

Keywords: Agentic AI, Retrieval-Augmented Generation, scholarship discovery, eligibility verification, citation grounding, deterministic fallback, safe abstention.

Introduction

Scholarship applicants routinely compare opportunities across institutions, countries, funding schemes, degree levels, fields of study, grade thresholds, work-experience requirements, language tests, and application deadlines. A conventional search engine may retrieve a relevant page, yet it does not reliably determine whether a particular applicant is eligible. A generative model may produce a fluent answer, but fluency does not guarantee that the answer is current, supported, or correctly cited.

ScholarAgent was developed to bridge this gap. It treats scholarship discovery as a structured, evidence-sensitive decision process rather than a free-form question-answering task. The system retrieves official scholarship records, evaluates a structured student profile, constructs source-linked claims, audits generated citations, and either returns a grounded result, recovers through deterministic fallback, or abstains when evidence is insufficient.

System scope

ScholarAgent is a preliminary screening assistant. It does not replace an official scholarship provider, admissions office, or funding decision. Every recommendation must be confirmed against the cited official source before an application is submitted.

Problem and Objectives

The project addresses three practical failure modes: retrieval without eligibility reasoning, generation without evidence control, and unsafe persistence after insufficient evidence or model failure. The implementation therefore aimed to:

- construct a provenance-preserving official scholarship corpus;
- compare BM25, dense, and hybrid retrieval;

- implement explainable deterministic eligibility screening;
- ground factual claims in stable evidence identifiers;
- verify citation integrity before accepting an answer;
- implement bounded query rewriting and answer repair with LangGraph;
- compare conventional single-pass RAG with Agentic RAG;
- evaluate fallback, abstention, runtime, and generation-call overhead;
- expose the workflow through a Streamlit demonstration.

The evaluation focused on retrieval quality, eligibility classification, citation safety, operational overhead, and safe behaviour under unsupported retrieval, expired deadlines, and local-model failure.

System Design

ScholarAgent uses a bounded state graph inspired by tool-using and reasoning-action agent patterns (Yao et al., 2023). Retrieval and generation attempts are explicitly limited so that every execution terminates in a grounded response, deterministic fallback, or abstention.

The architecture has four functional layers. The retrieval layer identifies candidate scholarships using lexical, semantic, or fused ranking. The eligibility layer evaluates structured applicant constraints. The grounding layer converts source fields into cited evidence and checks evidence sufficiency. The response layer generates, audits, repairs, falls back, or abstains according to explicit state and budget conditions.

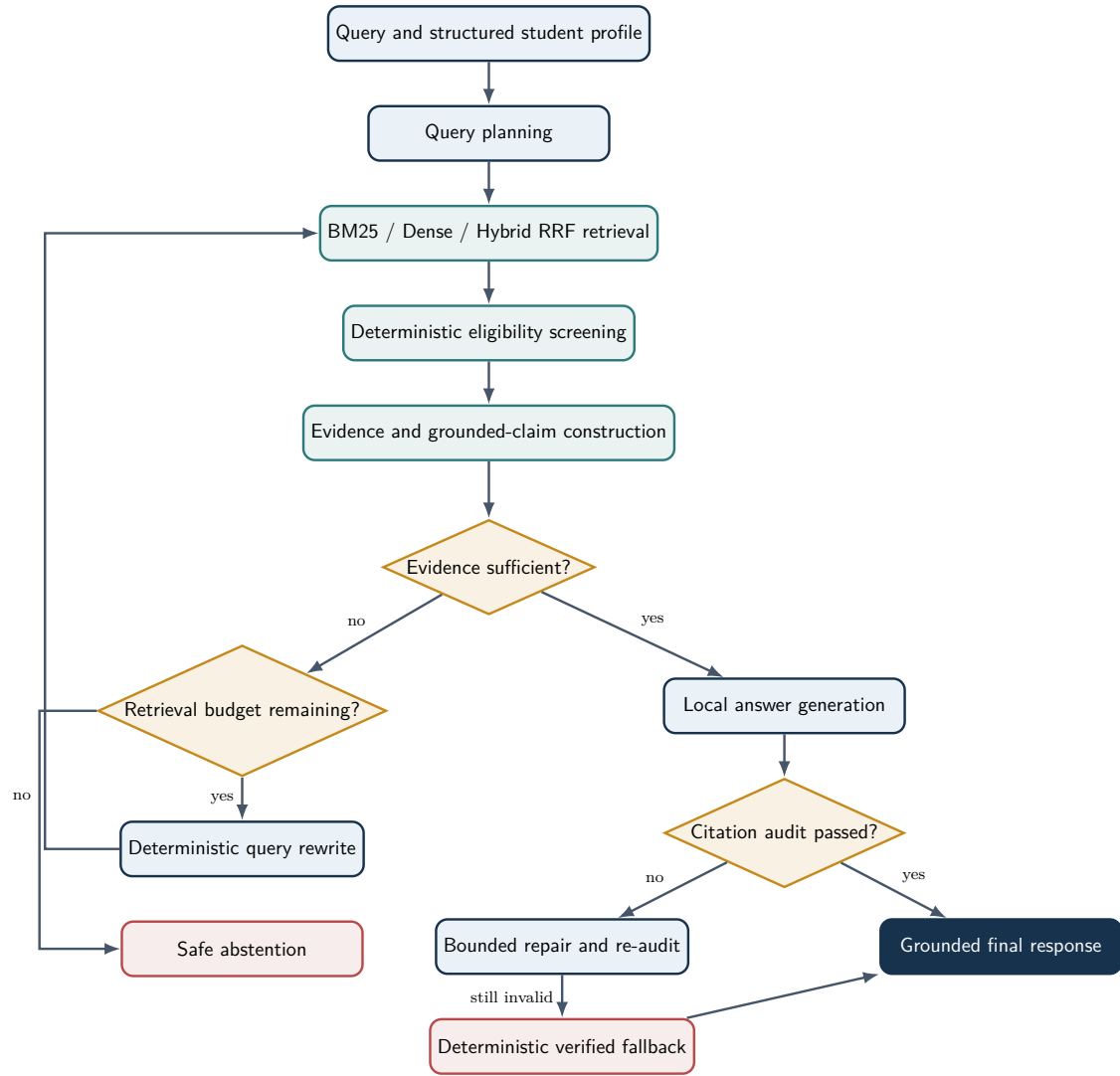


Figure 1: Bounded ScholarAgent workflow. The control logic separates retrieval and eligibility reasoning from generation, verification, and recovery.

Methodology

Retrieval strategies

BM25 provides the lexical baseline and follows the probabilistic relevance framework described by [Robertson and Zaragoza \(2009\)](#). Dense retrieval uses the local `nomic-embed-text:latest` Ollama model and cosine similarity. Hybrid retrieval combines lexical and semantic rankings through Reciprocal Rank Fusion ([Cormack et al., 2009](#)). The frozen hybrid configuration used a dense threshold of 0.60, `top_k=3`, `candidate_k=9`, an RRF constant of 60, and 768-dimensional embeddings.

Deterministic eligibility screening

The eligibility engine evaluates nationality, target degree, field of study, normalized GPA, work experience, language-test scores, application deadline, verified programme-specific requirements, and applicant preferences. It returns one of four statuses:

- **eligible:** no hard failure, missing field, or unresolved manual condition;
- **potentially_eligible:** structured rules pass but manual verification remains;

- `not_eligible`: at least one hard eligibility rule fails;
- `insufficient_information`: required applicant evidence is missing.

Expired rounds may remain retrievable for explanation, but the deadline rule prevents them from being presented as active recommendations.

Grounding and citation integrity

RAG combines retrieved context with generation (Lewis et al., 2020), but ScholarAgent adds deterministic controls around the generated text. Structured source fields become evidence snippets with stable identifiers. The verifier confirms that citation identifiers exist, claims cite evidence, evidence matches the structured record, official URLs are preserved, and no unexpected evidence is introduced. A failed answer receives one bounded repair attempt; if the repair remains invalid, the system constructs an extractive fallback answer from already verified claims. Unsupported queries abstain before generation.

Evaluation Protocol

The official corpus was divided into development, calibration, and held-out partitions, with no scholarship programme identity duplicated across partitions.

Table 1: Dataset partitions

Partition	Scholarship identities	Purpose
Development	3	Implementation and debugging
Calibration	6	Frozen threshold selection
Held-out	6	Final confirmatory evaluation
Total	15	Complete official-source corpus

The held-out benchmark contained 24 cases: 20 supported cases and four unsupported no-result cases. The four eligibility labels were balanced with five cases per class. The held-out programmes were Bristol Think Big, Glasgow Global Leadership, Aalto Excellence, Waikato Vice-Chancellor’s International Excellence, University of Manitoba Graduate Fellowship, and Maastricht NL-High Potential.

The RAG evaluation used `tinylama:latest` at temperature 0 on CPU. Retrieval and generation were each limited to two attempts. The transport timeout was set to 900 seconds because of CPU-only execution. No threshold, query, or prompt was tuned after inspecting held-out results.

Results

Retrieval performance

Table 2: Primary held-out retrieval results at $k = 3$

Retriever	P@3	R@3	MRR	Top-1	No-result
BM25	0.3333	1.0000	1.0000	1.0000	1.0000
Dense	0.3000	0.9000	0.5917	0.4000	1.0000
Hybrid RRF	0.3333	1.0000	0.8750	0.7500	1.0000

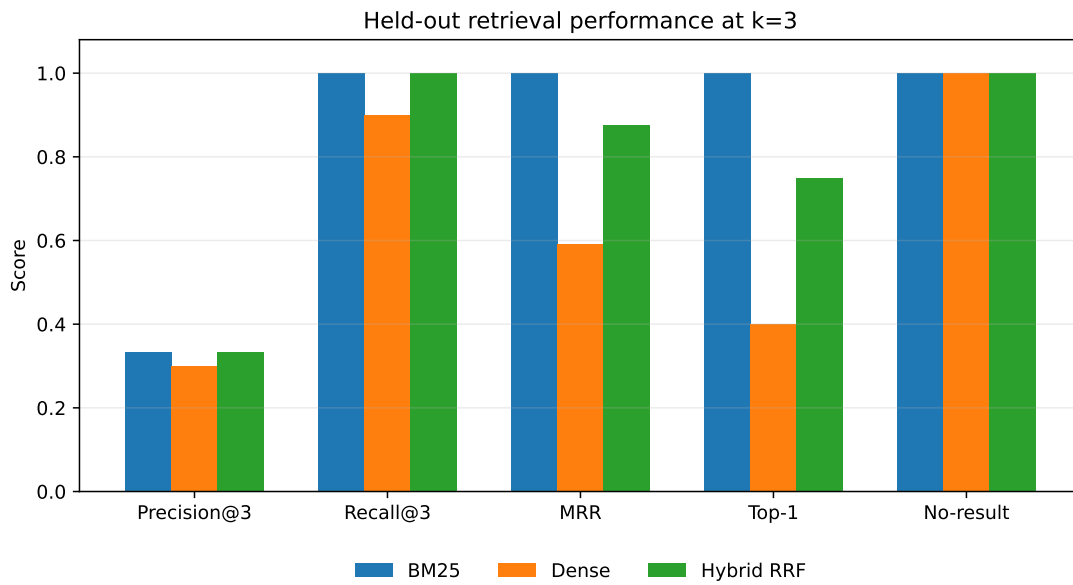


Figure 2: Held-out retrieval performance at $k = 3$.

BM25 performed best on this small corpus because benchmark queries contained lexically distinctive scholarship and university names. Dense retrieval showed weaker ranking, and hybrid fusion inherited some dense-ranking errors. This result is specific to the controlled corpus and should not be generalized to larger, more paraphrastic collections. A descriptive Recall@5 supplement produced recall values of 1.00 for BM25 and hybrid retrieval and 0.95 for dense retrieval; it was not used for tuning.

Eligibility classification

Table 3: Held-out eligibility classification

Status	Support	Precision	Recall	F1
Eligible	5	1.0000	1.0000	1.0000
Potentially eligible	5	1.0000	1.0000	1.0000
Not eligible	5	1.0000	1.0000	1.0000
Insufficient information	5	1.0000	1.0000	1.0000

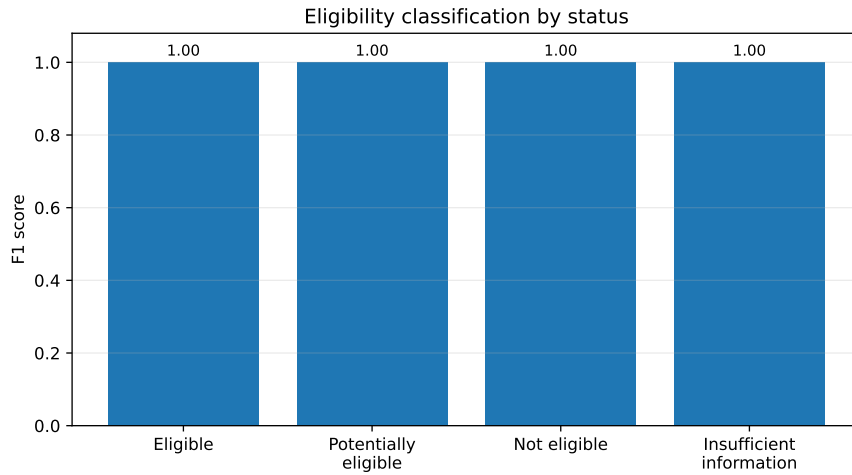


Figure 3: Eligibility F1 score by status.

Overall accuracy, macro precision, macro recall, macro F1, weighted F1, and no-result accuracy were all 1.00. The perfect result reflects a controlled benchmark aligned with the implemented rules and does not prove perfect performance on unrestricted scholarship policies.

Single-pass and Agentic RAG

Table 4: Single-pass and Agentic RAG comparison

Metric	Single-pass	Agentic
Positive completion rate	0.0000	1.0000
Positive citation-pass rate	0.0000	1.0000
Relevant grounding rate	0.9500	0.9500
Relevant citation rate	0.0500	0.7000
No-result accuracy	1.0000	1.0000
Mean latency (seconds)	140.251	293.064
Mean generation calls	0.8333	1.6667
Positive fallback rate	0.0000	1.0000

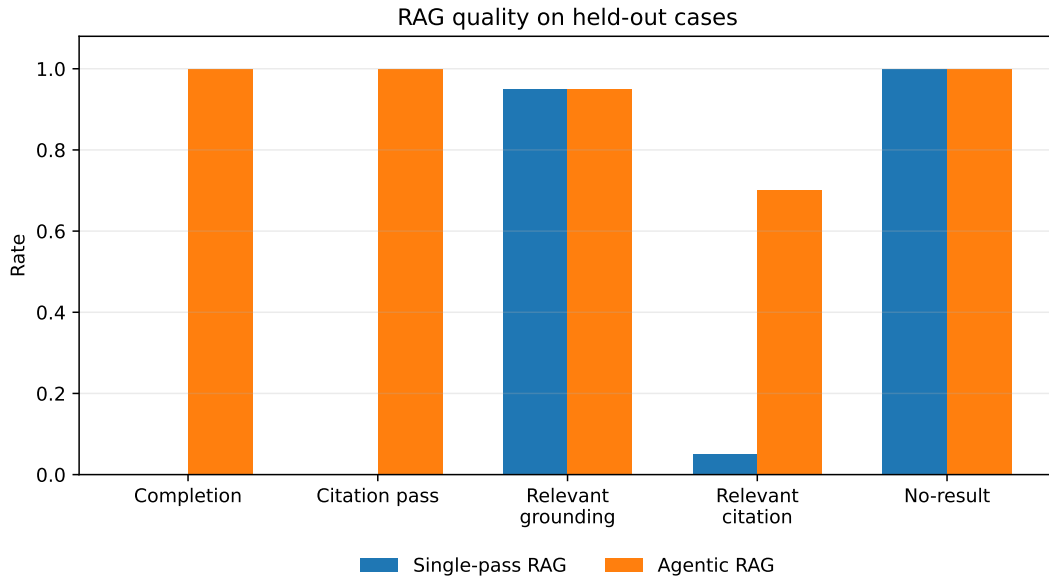


Figure 4: Quality comparison between single-pass RAG and the complete Agentic workflow.

Single-pass RAG grounded the relevant scholarship in 95% of positive cases but failed citation verification for every positive answer. Agentic RAG achieved complete citation-audit success through verification and deterministic fallback. The relevant-citation rate remained 0.70, showing that technical citation validity does not always guarantee citation of the benchmark’s intended scholarship.

Verification and operational overhead

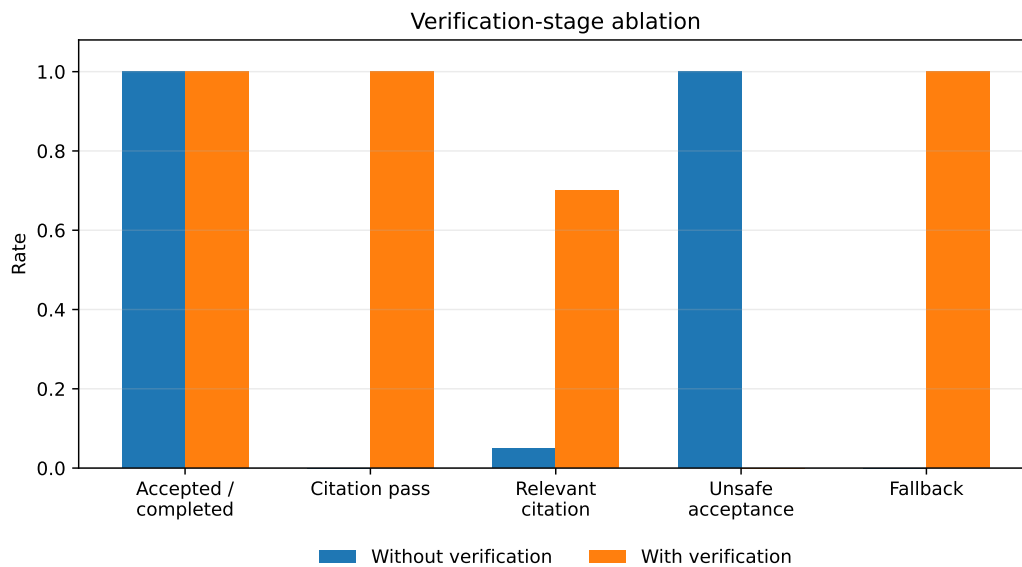


Figure 5: Counterfactual comparison with and without verification.

Without verification, all 20 positive first-pass answers would have been accepted even though none passed the post-hoc citation audit. With verification, all failures were detected. No TinyLlama citation repair succeeded, while all 20 cases were recovered through deterministic fallback. The safety benefit therefore came from explicit verification and recovery rather than successful language-model self-correction.

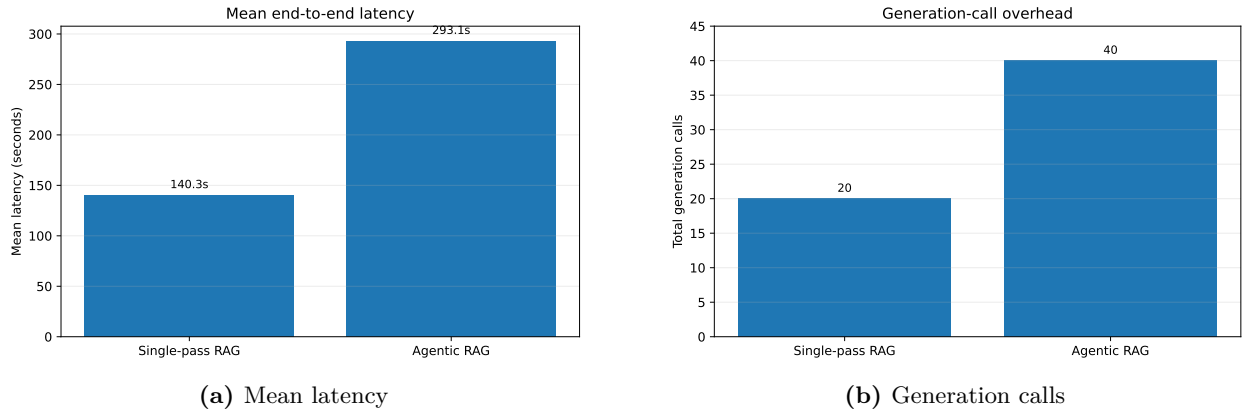


Figure 6: Operational overhead of the Agentic workflow.

Single-pass RAG required 56.101 total minutes and 20 generation calls. Agentic RAG required 117.226 total minutes and 40 generation calls, corresponding to approximately 2.09 times the total latency and twice the generation calls. Direct hosted API cost was zero because embeddings and generation ran locally through Ollama; local electricity, hardware use, CPU opportunity cost, and operator time were not monetized.

Interpretation of the main result

The evaluation does not show that TinyLlama learned to repair citations. It shows that deterministic verification prevented unsafe acceptance and that deterministic fallback recovered failed generations. ScholarAgent is therefore best described as a failure-aware agent architecture with explicit safety controls.

Demonstration Application

A Streamlit interface exposes the system through structured profile entry, BM25 or hybrid retrieval, eligibility screening, official evidence inspection, citation verification, fallback status, and safe abstention.

The screenshot shows the ScholarAgent web interface. On the left is a 'Demo configuration' sidebar with settings for execution mode (Fast verified demo), retriever (BM25-only offline mode), corpus (official development corpus), embedding (openai:embed-text:latest), generator (tinyllama:latest), dense threshold (0.5), and top-k (3). The main area is titled 'ScholarAgent' and 'Evidence-grounded scholarship discovery and eligibility verification'. It contains a '1. Scholarship query' section with an 'Information need' field containing 'erasmus mundus scholarship'. Below this is a '2. Student profile' section with various input fields: Nationality (Nigerian), Fields of study (Artificial Intelligence, Data Science), Country of residence (Finland), Preferred countries (Sweden, Netherlands), Target degree (master), Years of work experience (2.00), GPA (4.20), GPA scale (5.00), and Language test (IELTS). There are checkboxes for 'Requires full funding', 'Include GPA', and 'Include language score'. An 'Assessment date' field shows '2026/10/15'. At the bottom is a blue 'Run ScholarAgent' button.

Figure 7: ScholarAgent query and structured student-profile interface.

The recommended presentation configuration uses *Fast verified demo* with *BM25-only offline mode*. This configuration performs zero external language-model calls while demonstrating the real verification and deterministic fallback pathway.

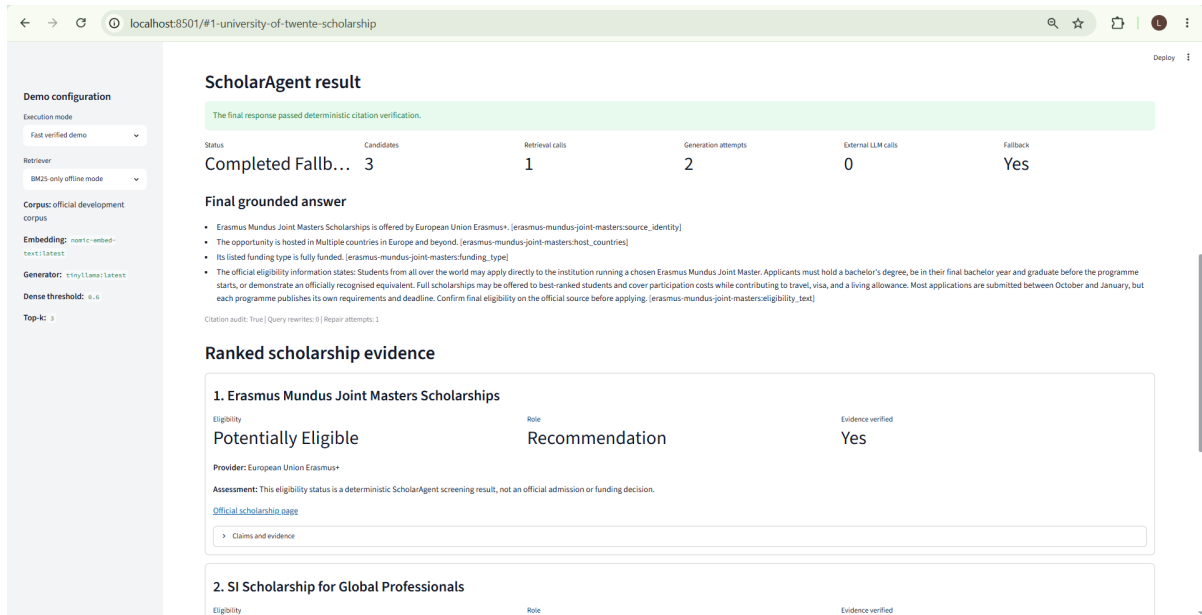


Figure 8: Verified fallback result with ranked evidence, official source links, citations, and execution counters.

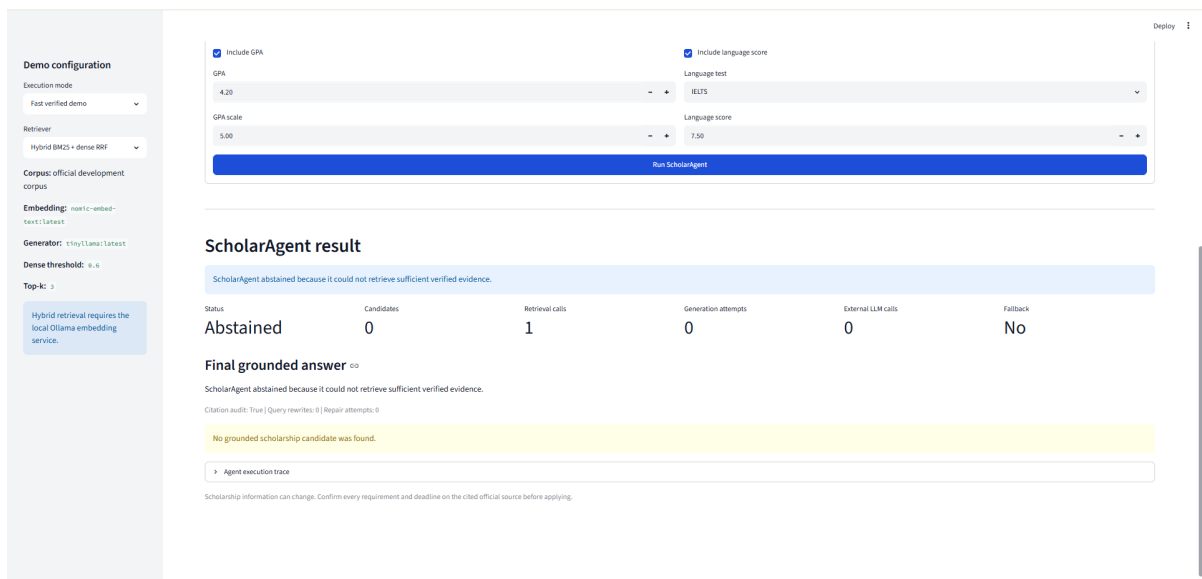


Figure 9: Safe abstention after unsupported retrieval: one retrieval call, zero generation calls, and no fallback.

Discussion

The deterministic eligibility engine provided transparent and reproducible decisions for structured requirements. The evaluation pipeline froze calibration settings before held-out testing, preserved official-source provenance, and reported both primary and post-hoc metrics. The graph handled unsupported retrieval, expired deadlines, citation failure, and transport timeout without silently returning an unsupported recommendation.

The principal weakness was the local generator. TinyLlama produced no citation-safe positive answer and no successful repair. Dense retrieval also underperformed the lexical baseline on the small, lexically distinctive corpus. Query rewriting was not triggered on the held-out benchmark, so the final evaluation provides no evidence that rewriting improved performance.

Verification eliminated unsafe first-pass acceptance but approximately doubled runtime and generation calls. For high-consequence recommendation systems this trade-off may be acceptable, but production deployment would require faster inference, stronger ranking, and a more reliable instruction-following model.

Limitations and Responsible Use

The evaluation is intentionally controlled and has important limitations:

- the held-out corpus contains six scholarship identities and 24 cases;
- official pages are represented as dated snapshots;
- deterministic rules do not cover every natural-language policy;
- the embedding model was not adapted to the scholarship domain;
- every successful positive Agentic output used deterministic fallback;
- benchmark-relevant citation remained 0.70;
- local CPU inference produced high latency;
- direct API cost does not represent total local computation cost.

A responsible deployment should display official source links and last-checked dates, refresh source records regularly, protect applicant profiles, support human review for ambiguous requirements, retain safe abstention, and clearly state that eligibility results are preliminary screening outcomes.

Future Work

The repository contains 190 passing automated tests covering schemas, corpus validation, retrieval, eligibility rules, deadline handling, grounding, citation auditing, single-pass RAG, Agentic RAG, query rewriting, fallback, abstention, evaluation artifacts, report assets, and the demonstration interface.

Future work should expand the official corpus, create a larger independently annotated benchmark, automate source refresh and change detection, evaluate domain-adapted embeddings, test a stronger instruction-following generator, add human citation-correctness evaluation, measure local energy and hardware costs, and deploy a public deterministic demonstration that does not depend on a local Ollama service.

Conclusion

ScholarAgent demonstrates a bounded, evidence-grounded, and failure-aware Agentic RAG architecture for scholarship discovery and eligibility screening. The project achieved strong controlled-benchmark retrieval and eligibility performance, reliable unsupported-query abstention, explicit expired-opportunity handling, reproducible evaluation, and a functional Streamlit interface.

The central finding is that verification materially changed system safety. Every positive first-pass language-model answer failed citation verification. Deterministic verification and fallback converted those failures into citation-safe completions, although at substantial runtime cost and without fully solving benchmark-relevant citation selection. The results support a hybrid architecture in which language models assist interaction and synthesis while deterministic components enforce eligibility, provenance, citation integrity, bounded execution, and safe failure.

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